



Machine Learning

Training Data vs Test Data

A fun lesson for young learners

Grade 5 | AI & Digital Skills

What is Artificial Intelligence?

AI helps machines think and learn like humans!

- AI is everywhere around us
- It's in your phone and games
- Robots use AI too!

What is Machine Learning?

Machine Learning means teaching computers to learn from **data** !

- Computers find patterns in information
- They improve by practicing over time
- No one tells them the exact answers

Why Do Machines Need Data?

Data is information that helps machines learn.

- More data helps machines learn better
- It's like a recipe — more ingredients, better results!
- Machines need lots of examples

Training Data — Teaching the Machine

Training data is information used to **teach** the machine.

- Apple → "This is an apple"
- Banana → "This is a banana"
- Mango → "This is a mango"

Test Data — Checking the Machine

Test data is new information to check if the machine **learned well** . It's like a surprise quiz!

- Show new fruit pictures
- No labels given
- Can the machine guess correctly?

Training Data

- Used to teach machines

Test Data

- Used to check learning

- Has labels or answers
- Machines learn patterns here
- Like a textbook with answers

- No labels shown to machine
- Tests if machine predicts right
- Like a surprise quiz!

🔗 How Do Machines Learn From Data?

- **Step 1:** Look at training data
- **Step 2:** Find patterns and rules
- **Step 3:** Make guesses on new data
- **Step 4:** Check if guesses are correct!

🍎 Example: Recognizing Fruits

The machine learns fruit names by looking at many pictures!

- Training: labeled photos of fruits
- Machine learns: "Red and round = Apple!"
- Practice makes perfect!

Quiz Time!

Answer these 20 questions about Machine Learning. Circle the correct option!

Q1.

What does AI stand for?

- A) Automatic Internet
- B) Artificial Intelligence
- C) Active Input
- D) Advanced Information

Q2.

AI helps machines to...

- A) Sleep and rest
- B) Think and learn like humans
- C) Cook food
- D) Grow plants

Q3.

Where can you find AI?

- A) Only in space
- B) In your phone and games
- C) Only in books
- D) Nowhere near us

Q4.

What does Machine Learning mean?

- A) Teaching computers to learn from data
- B) Fixing broken machines
- C) Building robots by hand
- D) Painting computers

Q5.

How do computers get smarter in Machine Learning?

- A) By sleeping
- B) By practicing over time
- C) By staying off
- D) By guessing randomly forever

Q6.

In Machine Learning, who gives the machine the exact answers?

- A) A teacher every time
- B) No one — it finds patterns itself
- C) A robot friend
- D) The internet

Q7.

What is data?

- A) A type of robot
- B) Information that helps machines learn
- C) A computer game
- D) A kind of fruit

Q8.

More data usually helps a machine...

- A) Learn worse
- B) Learn better
- C) Stop working
- D) Forget everything

Q9.

The lesson compares data to a recipe. What does more ingredients mean?

- A) Worse results
- B) Better results
- C) No results
- D) Slower cooking

Q10.

What is training data used for?

- A) To teach the machine
- B) To confuse the machine
- C) To delete data
- D) To turn off the computer

Q11.

In the fruit example, what does training data include?

- A) Unlabeled pictures
- B) Labeled pictures like "This is an apple"
- C) Empty pictures
- D) Videos only

Q12.

What is test data used for?

- A) To check if the machine learned well
- B) To teach the machine from scratch
- C) To delete the training data
- D) To charge the computer

Q13.

Test data is like a...

- A) Surprise quiz
- B) Nap time
- C) Birthday party
- D) Recipe book

Q14.

Does test data come with labels for the machine?

- A) Yes, always
- B) No labels are given
- C) Only sometimes on weekends
- D) Labels are optional for the machine to see

Q15.

Which one is TRUE about training data?

- A) It has labels or answers
- B) It is always test data
- C) It is never used
- D) It has no purpose

Q16.

Which one is TRUE about test data?

- A) It teaches patterns to the machine
- B) It checks if the machine predicts correctly
- C) It is shown with labels
- D) It is the same as training data

Q17.

What is Step 1 in how machines learn from data?

- A) Make guesses on new data
- B) Check if guesses are correct
- C) Look at training data
- D) Find patterns and rules

Q18.

What is the last step in how machines learn from data?

- A) Look at training data
- B) Find patterns and rules
- C) Make guesses on new data
- D) Check if guesses are correct

Q19.

In the fruit example, what rule might the machine learn?

- A) "Red and round = Apple!"
- B) "Green means banana"
- C) "All fruits are the same"
- D) "Fruits cannot be learned"

Q20.

Training data is like a textbook with answers. Test data is like a...

- A) Blank notebook
- B) Surprise quiz
- C) Storybook
- D) Calendar

✓ Answer Key

Q1

B

Q2

B

Q3

B

Q4

A

Q5

B

Q6

B

Q7

B

Q8

B

Q9

B

Q10

A

Q11

B

Q12

A

Q13

A

Q14

B

Q15

A

Q16

B

Q17

C

Q18

D

Q19

A

Q20

B